

Experiment 1: Context-Free Grammar Rule Validation and Parse Tree Construction

Program:

```
Det = ["the", "a"]
```

```
N = ["student", "teacher", "book"]
```

```
V = ["reads", "likes"]
```

```
def parse_cfg(sentence):
```

```
    words = sentence.split()
```

```
    if len(words) != 5:
```

```
        return "Invalid Sentence"
```

```
    if (words[0] in Det and
```

```
        words[1] in N and
```

```
        words[2] in V and
```

```
        words[3] in Det and
```

```
        words[4] in N):
```

```
        parse_tree = {
```

```
            "S": {
```

```
                "NP": {
```

```
                    "Det": words[0],
```

```
                    "N": words[1]
```

```
                },
```

```
                "VP": {
```

```
                    "V": words[2],
```

```
                    "NP": {
```

```
                        "Det": words[3],
```

```
                        "N": words[4]
```

```
                    }
```

```

        }

    }

}

return parse_tree

else:

    return "Invalid Sentence"

sentences = [

    "the student reads a book",

    "a teacher likes the book",

    "student reads book",

    "the book likes a teacher",

    "reads the student book"

]

for s in sentences:

    print("\nInput:", s)

    print("Output:", parse_cfg(s))

```

OUTPUT:

The screenshot shows two side-by-side IDE windows. The left window, titled 'IDLE Shell 3.12.10', displays the output of the Python script. It shows the program restarting and processing five sentences. For 'the student reads a book', it outputs a detailed parse tree. For 'a teacher likes the book', it also outputs a parse tree. For 'student reads book', 'the book likes a teacher', and 'reads the student book', it outputs 'Invalid Sentence'.

The right window, titled 'co3 at1.py - C:/Users/vdmoh/Downloads/co3 at1.py (3.12.10)', shows the source code of the script. It defines a list of sentences, a function `parse_cfg` that checks for valid sentence structures and returns a parse tree or 'Invalid Sentence', and a loop that prints the input and output for each sentence.

```

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/vdmoh/Downloads/co3 at1.py =====
Input: the student reads a book
Output: {'S': {'NP': {'Det': 'the', 'N': 'student'}, 'VP': {'V': 'reads', 'NP': {'Det': 'a', 'N': 'book'}}}}
Input: a teacher likes the book
Output: {'S': {'NP': {'Det': 'a', 'N': 'teacher'}, 'VP': {'V': 'likes', 'NP': {'Det': 'the', 'N': 'book'}}}}
Input: student reads book
Output: Invalid Sentence
Input: the book likes a teacher
Output: {'S': {'NP': {'Det': 'the', 'N': 'book'}, 'VP': {'V': 'likes', 'NP': {'Det': 'a', 'N': 'teacher'}}}}
Input: reads the student book
Output: Invalid Sentence
>>>

co3 at1.py - C:/Users/vdmoh/Downloads/co3 at1.py (3.12.10)
File Edit Format Run Options Window Help
v = ["reads", "likes"]

def parse_cfg(sentence):
    words = sentence.split()
    if len(words) != 5:
        return "Invalid Sentence"

    if (words[0] in Det and
        words[1] in N and
        words[2] in V and
        words[3] in Det and
        words[4] in N):

        parse_tree = {
            "S": {
                "NP": {
                    "Det": words[0],
                    "N": words[1]
                },
                "VP": {
                    "V": words[2],
                    "NP": {
                        "Det": words[3],
                        "N": words[4]
                    }
                }
            }
        }

        return parse_tree

    else:
        return "Invalid Sentence"

sentences = [
    "the student reads a book",
    "a teacher likes the book",
    "student reads book",
    "the book likes a teacher",
    "reads the student book"
]

for s in sentences:
    print("\nInput:", s)
    print("Output:", parse_cfg(s))

```

Experiment 2: Agreement and Feature Structure Checking

Program:

```
subjects = {  
    "he": ("singular", "third"),  
    "she": ("singular", "third"),  
    "it": ("singular", "third"),  
    "they": ("plural", "third")  
}  
  
verbs = {  
    "runs": "singular",  
    "writes": "singular",  
    "run": "plural",  
    "write": "plural"  
}  
  
def check_agreement(subject, verb):  
    if subject not in subjects or verb not in verbs:  
        return False  
  
    subject_number = subjects[subject][0]  
    verb_number = verbs[verb]  
  
    return subject_number == verb_number  
  
test_cases = [  
    ("he", "runs"),  
    ("he", "run"),
```

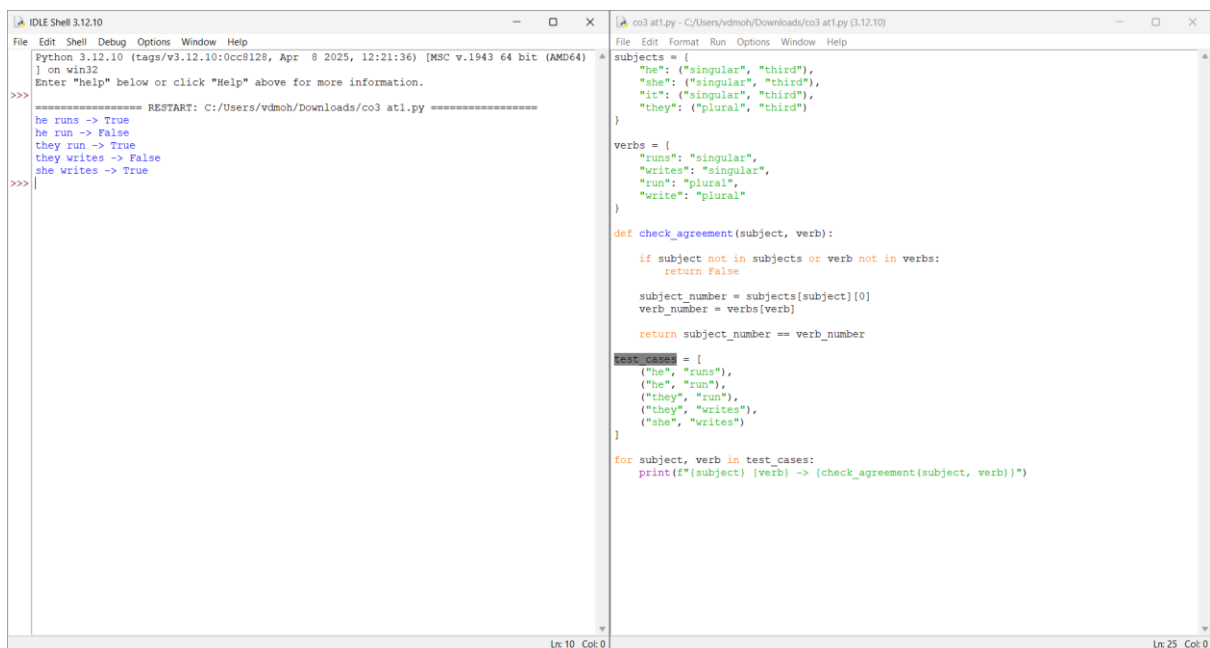
```
("they", "run"),  
("they", "writes"),  
("she", "writes")
```

```
]
```

```
for subject, verb in test_cases:
```

```
    print(f'{subject} {verb} -> {check_agreement(subject, verb)}')
```

OUTPUT:



The screenshot shows a Python IDLE window with two panes. The left pane displays the output of a script, and the right pane displays the source code of the script.

Left Pane (Output):

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)]  
 on win32  
Enter "help" below or click "Help" above for more information.  
>>>  
===== RESTART: C:/Users/vdmoh/Downloads/co3 at1.py =====  
he runs -> True  
he run -> False  
they run -> True  
they writes -> False  
she writes -> True  
>>>
```

Right Pane (Source Code):

```
co3 at1.py - C:/Users/vdmoh/Downloads/co3 at1.py (3.12.10)  
File Edit Format Run Options Window Help  
subjects = [  
    "he": ("singular", "third"),  
    "she": ("singular", "third"),  
    "it": ("singular", "third"),  
    "they": ("plural", "third")  
]  
  
verbs = [  
    "runs": "singular",  
    "writes": "singular",  
    "run": "plural",  
    "write": "plural"  
]  
  
def check_agreement(subject, verb):  
    if subject not in subjects or verb not in verbs:  
        return False  
  
    subject_number = subjects[subject][0]  
    verb_number = verbs[verb]  
  
    return subject_number == verb_number  
  
test_cases = [  
    ("he", "runs"),  
    ("he", "run"),  
    ("they", "run"),  
    ("they", "writes"),  
    ("she", "writes")  
]  
  
for subject, verb in test_cases:  
    print(f'{subject} {verb} -> {check_agreement(subject, verb)}')
```

Experiment 3: Probabilistic Context-Free Grammar (PCFG) Parsing

PROGRAM:

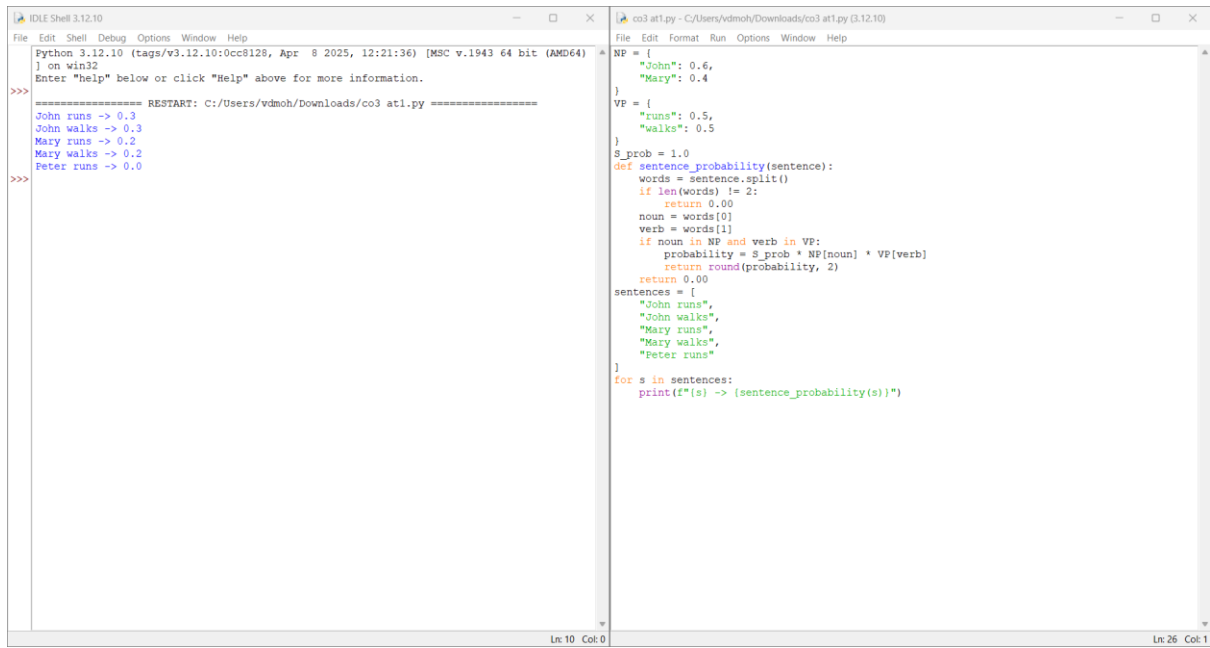
```
NP = {  
    "John": 0.6,  
    "Mary": 0.4  
}  
  
VP = {  
    "runs": 0.5,  
    "walks": 0.5  
}  
  
S_prob = 1.0  
  
def sentence_probability(sentence):  
    words = sentence.split()  
  
    if len(words) != 2:  
        return 0.00  
  
    noun = words[0]  
    verb = words[1]  
  
    if noun in NP and verb in VP:  
        probability = S_prob * NP[noun] * VP[verb]  
        return round(probability, 2)  
  
    return 0.00  
  
sentences = [  
    "John runs",  
    "John walks",  
    "Mary runs",  
    "Mary walks",  
    "Peter runs"
```

]

for s in sentences:

```
print(f"{s} -> {sentence_probability(s)}")
```

OUTPUT:



The image shows two side-by-side Python IDLE windows. The left window, titled 'Python 3.12.10', displays the output of a script. The right window, titled 'co3 at1.py', displays the source code of the script.

Left Window Output:

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/vdmoh/Downloads/co3 at1.py =====
John runs -> 0.3
John walks -> 0.3
Mary runs -> 0.2
Mary walks -> 0.2
Peter runs -> 0.0
>>>
```

Right Window Code:

```
File Edit Format Run Options Window Help
NP = {
    "John": 0.6,
    "Mary": 0.4
}
VP = {
    "runs": 0.5,
    "walks": 0.5
}
S_prob = 1.0
def sentence_probability(sentence):
    words = sentence.split()
    if len(words) != 2:
        return 0.00
    noun = words[0]
    verb = words[1]
    if noun in NP and verb in VP:
        probability = S_prob * NP[noun] * VP[verb]
        return round(probability, 2)
    return 0.00
sentences = [
    "John runs",
    "John walks",
    "Mary runs",
    "Mary walks",
    "Peter runs"
]
for s in sentences:
    print(f"{s} -> {sentence_probability(s)}")
```